

Designing Software for Genomics Medicine Service Leaders to Engage Stakeholders

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Abstract— *Background:* Stakeholder engagement can be challenging yet is critically important for the success of a new genomic medicine service or genetic test offering. Services may include ordering, analysis, interpretation, and counseling associated with using a patient's genetic information to make decisions about medical choices. *Objective:* The aim of this study was to use a human-centered design (HCD) approach to design software for genomic medicine service leaders seeking to engage stakeholders. Of particular interest were stakeholders that could be partners to champion a new genetic test offering. *Methods:* Our HCD process was driven by a modified design sprint methodology involving participatory design sessions and semi-structured interviews with experts. Subsequently, we created a low-fidelity prototype aiming to facilitate early engagement with potential genomic medicine service partners. The prototype was used to test the design with current genomic medicine service leaders. To guide our evaluation, we developed a set of design considerations for the software application that reflected common strategies used by programs successfully implementing genetic test offerings in diverse settings. *Results:* We analyzed notes collected from interview sessions with seven genomic medicine service leaders. We identified ten sub-themes, synthesized corresponding notes, and focused our interpretations around three design considerations: #1-identify potential genomic medicine service partners to champion a new genetic test offering; #2-train and educate stakeholders; and #3-obtain and use genomic medicine partner feedback. The top three ranked sub-themes were: 1 - add new information, 2 - general approval, and 3 - add/change functionality. *Conclusion:* Our findings suggest that genomic medicine service leaders approve of our software design and process to facilitate key stakeholders' review and endorsement of a new genetic test offering. We also demonstrated an evaluation strategy that draws from lessons of successful genomic medicine programs to identify and prioritize areas to improve the software in future design iterations.

Keywords—human-centered design; software design; genomic medicine; stakeholder engagement; organizational readiness

I. INTRODUCTION

Despite the promise of genomic medicine, there remains a translational gap. Most of the funding and research in genomic medicine is focused on the initial stages of discovery, while very few proceed to the implementation and evaluation phases. It is estimated that around 97% of published research focuses on discovery research or developing a candidate test or therapy, while no more than 3% of published research focuses on candidate application evaluation, policy development, recommendation implementation, and health outcomes assessment [1], [2]. This "Lost in Translation" problem not only results in few tests being implemented but may further stifle innovation and discourage the research community as discoveries are rarely brought to the clinic [3].

To better facilitate the success of evidence-based genomic medicine practices for common diseases, genomic medicine service teams are emerging at some institutions to provide support for ordering, analyzing, interpreting, and counseling associated with using patients' genetic information to make decisions about medication choices. While these service teams exist at a growing number of institutions, barriers remain in the clinical adoption of services provided. Major challenges for implementation include limited evidence and conflicting interpretation of benefits, limited access to genomic medicine expertise and testing, lack of institutional and clinician acceptance, etc. [4].

Stakeholder engagement is cited as a factor that is important for the success of genomic medicine services [5], [6]. To bring genomic medicine services and genetic test offerings into clinical and public health practices, their evidence, risks, and benefits should be cautiously assessed to distinguish genomic

applications that can improve health from those that might result in potential harm and unnecessary healthcare expenditure [3]. A formal and organized evaluation process that engages stakeholders is not only crucial in assisting with such goal, but also instrumental in helping gain buy-in from stakeholders, ensuring that the various parties involved reach consensus and are committed to the whole translation process [7].

In this paper, we describe the design of the “review and prioritization” software tool intended to help genomic medicine service leaders to engage stakeholders. In section II, we provide information about our design motivation; In section III we give a detailed description of our methodology; In section IV we describe the results from our research activities, including an overview of the low-fidelity prototype for user testing; and in section V, we give a discussion of the potential impact of this work, the limitations, and areas for future work.

II. DESIGN MOTIVATION

The primary design motivation was to enable stakeholder engagement around deciding what types of genomic medicine services and genetic test offerings to support at an institution. If we consider implementing genomic medicine services more broadly as a means for quality improvement, the importance of clinician stakeholder involvement is well-recognized [8]. Clinical champions with relevant experiences can help other healthcare providers become aware of genomic medicine services and genetic test offerings, and of how they might be used in their practice [9]. Such clinical stakeholders can provide a positive perception of the services and options, identify points of potential resistance, and leverage their influence to overcome resistance in clinical settings given their frequent interactions with other clinical professionals on the frontline [10]. Potential reasons for a lack of engagement include a perception of the cost of change exceeding the anticipated benefit [11] and a conflict in leadership priorities or values set by physicians [12]. Furthermore, disengagement can occur due to a non-supportive practice culture or organizational structure [13].

Organizational readiness for change is one mechanism by which stakeholder engagement may lead to an improved likelihood of a successful genomic medicine service. Organizational readiness refers to the organizational members’ level of commitment to change and ability to implement the change [14]. According to the theory on organizational readiness, quality improvement is most likely to occur when individuals feel committed to the change and perceive that they will have the ability to implement that change [14].

Drawing from that theory, there are important considerations for the design of software to enable stakeholder engagement: (1) facilitating key stakeholder review and endorsement prior to pursuing a new genomic medicine service or genetic test offering can improve the likelihood of its success, and (2) seeking the opinions and feedback of key stakeholders on genomic medicine services and genetic test offerings early in the process can lead to those stakeholders becoming *service partners* that are both committed to it and perceive it can be successfully implemented at their organization.

III. METHODS

Our human-centered design process was driven by a modified design sprint methodology [16]. Main research activities included participatory design sessions, semi-structured interviews with experts on institutional review and approval processes for new genetic test offerings provided by a genomic medicine (GM) service team, and end-user testing sessions with institutional leaders that propose and implement such offerings (i.e., *GM service leaders*).

A. Participatory Design Sessions

From mid-October 2022 to mid-January 2023, we held design sessions in which we integrated a modified design sprint methodology. The design sprint process includes a series of user-centered design activities and assigns a specific schedule to complete those activities in five consecutive days [16]. Similar to others seeking to apply this methodology in a research setting [17], we made several adaptations. In summary, we scheduled specific periods for team members to attend design sessions and added time to organize, reflect on, and verify findings from expert consultations. We also delayed end-user testing to establish and internally test a semi-structured interview guide. Fig. 1 illustrates the main design sprint steps including: (1) mapping out paths to a solution, (2) sketching potential new solutions, (3) deciding which sketched solutions to pursue further and creating a storyboard, (4) turning the storyboard into a prototype, and (5) testing the prototype with end users.

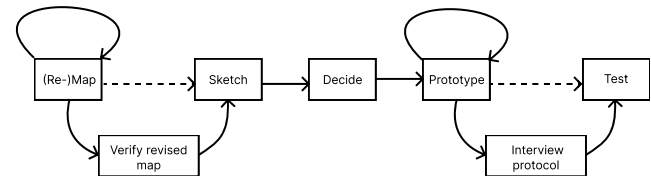


Fig. 1. Overview of modified design sprint steps.

We used participatory design sessions to share empirical findings from expert consultations at the first mapping step and following the fifth end-user testing step. After the mapping step, during our session, we spent time reflecting on findings from expert consultations to revise our map and verify the map prior to the next stage of the design process. Following the prototyping step, we also spent time creating and iterating on the prototype and a semi-structured interview protocol prior to testing.

B. User Role and System Functionality

The “review & prioritization” software was meant to be used by *GM service leaders*, and by potential *GM service partners*. The role of a *GM service leader* is to assist with the ordering, analysis, interpretation, and counseling associated with using a patient’s genetic information to make clinical decisions. Someone in this role would also select specific genetic test offerings to support on an organizational level. The review & prioritization software is intended for use on a regular basis to manage a list of genetic test offerings being considered at their institution. A *GM service partner* is a key stakeholder that is committed to supporting and promoting the use of a particular genetic test offering. These individuals may include clinical leaders and clinical staff that would be impacted by the proposed

offering and whose engagement is important for its adoption. The focus of this paper is on describing feedback from GM service leaders.

C. Building Prototype System Design and Functionality

Figma (figma.com) was used to design and build a low-fidelity, clickable prototype based on the core system functionalities and the role of potential users. We focused our content examples on genetic test offerings to bring pharmacogenomics-guided drug therapy into use clinically. In this context, pharmacogenomics can be used as a clinical tool to assist in drug selection and dosing and reduce the risk for drug-related problems.

Pharmacogenomic (PGx) tests are particularly relevant because they may require more strategies to engage stakeholders than disease-focused genetic tests due to a need for more extensive infrastructure to integrate into routine care [6]. Content presented in the prototype included three metrics described in the literature as important areas to provide evidence around pharmacogenomic testing: drug-related problems, economic value, and start-up effort. We included *drug-related problems* as a metric given recognition from groups such as the Federal Drug Administration that adverse drug reactions can be prevented with pre-emptive genetic testing [18]. *Economic value* was included given the importance of having accurate estimates of costs and use of health services [19]. *Start-up effort* was included in recognition that there can be pre-requisites for the adoption of a genetic test e.g., improving genetic test turnaround time [20].

To create role-specific semi-structured interview protocols, end-user tasks and content-related questions were initially created by JCB, ZL, KY and piloted with the entire team during a participatory design session. Following that session, revisions to the prototype and the interview protocol were made.

D. End-User Testing Procedure

GM service leaders were interviewed and asked to complete role-specific tasks. They were provided with a brief introduction to the test's purpose and process and given minimal background on the review & prioritization software. The end-user testing sessions were conducted remotely on Zoom by one interviewer (JCB). Three researchers from the team also collected field notes throughout each session. Each GM service leader was asked a few pre-interview questions related to their knowledge of genetics and experience with institutional processes to review and prioritize genomics applications for clinical use. Participants were then asked to think aloud while carrying out the end-user tasks. Following each task, participants were also asked questions about the content presented to them.

At the end of the session, the participants completed a brief semi-structured interview to capture thoughts that were not articulated during the session. Overall, the sessions were intended to last 45-60 minutes in duration. To allow for better recall at the data analysis stage, after each session we created a summary of the main points, key findings, and the repetition of keywords, ideas, and phrases used by the participant.

E. Evaluation Guided by Design Considerations

We analyzed notes taken during end-user sessions for themes. After we completed data collection for each end user group, we used a thematic data analysis approach [21], [22] to analyze the notes collected under three main topics: issues/challenges, recommendations, and likes. Notes under each task were assigned initial codes for themes and sub-themes by one of two reviewers, and a second reviewer indicated agreement or disagreement with each code. Disagreements were resolved during group discussions and consensus was reached on the final coded themes and sub-themes. We maintained a detailed description of the user feedback gathered from the notes for the sub-theme assignments.

A final count of the unique sub-themes associated with user feedback from the user testing session was calculated. To synthesize our results, we used an online whiteboard called Miro (Miro.com) [23] to create affinity maps for each sub-theme. The affinity mapping allowed us to organize, define and summarize details covered by each sub-theme in a visual way and to quickly assess the overall details of each sub-theme. The outcome of this process was a set of "synthesized comments" for each sub-theme.

To prioritize areas to improve the software in future design iterations, we aligned sub-themes and synthesized comments with design considerations informed by successful programs offering genetic testing services. Despite the diversity that exists in successful programs to implement such services with respect to program goals, approaches, and settings [5], [6], there are commonalities that exist in their implementation strategies. These included: identifying individuals whose engagement is important for the success of a new service (i.e., key stakeholders); training and educating stakeholders; and obtaining and using stakeholder feedback throughout implementation [6]. Thus, we specified design considerations where these success factors aligned well with functionalities supported by our design. Lastly, we described insights gained from our review of synthesized comments about functionality supporting the three design considerations.

IV. RESULTS

A. Map Following Expert Consultations

For our initial map, we sought to describe the interface between various oversight review committees responsible for the approval of a new genetic test offering. We thus interviewed four experts that currently or previously participated in clinical decision support (CDS) review committees and/or pharmacy and therapeutic committees. Findings from those consultations clarified that process and allowed us to revise our map to better reflect current processes toward offering a new genetic test. In addition, all four experts suggested stakeholder engagement as an upstream requirement to introducing a genetic test offering to any institutional oversight committee. Our design goal thus focused on that need.

B. Prototype Software Functionality and End-user Tasks

Overall, we defined 16 tasks for the two user processes intended for GM service leaders. We also had two onboarding tasks for the prototype that were excluded from our data analyses.

Software functionality explored for each user process is summarized below:

- **User Process #1: Create and distribute a review form to potential GM service partners.** The envisioned software provides (1) a means for a GM service leader to create a *review form* to collect feedback on the proposed genetic testing, and (2) a means for a GM service leader to distribute the review form (with a description of a proposed genetic testing) to potential GM service partners. Upon distributing a review form, an email invitation to complete the form would be sent to specified individuals. We defined 9 tasks for end users to review this functionality.
- **User Process #2: Act upon review form responses.** The envisioned software provides a means for a GM service leader to view and act upon review form responses. We defined 6 tasks for end users to review this functionality.

C. Prototype Design Considerations

We proposed three design considerations where success factors of programs offering genetic testing services align well with functionalities supported by our design.

- **Design consideration #1: Identify potential GM service partners to champion a new genetic test offering.** Upon creating a genetic test review form, the system user can search for and select potential GM service partners for distribution. A screenshot of the prototype functionality is shown in Fig. 2. [User Process #1, Task #7 – Select clinical expert]
- **Design consideration #2: Train and educate stakeholders.** Our design of the genetic test review form currently includes three default metrics that can be customized by the system user prior to distributing to GM service partners. Metrics included drug-related problems, economic value, and start-up effort. Fig. 3 illustrates sample content displayed in the review form when proposing *CYP2C19* genetic testing prior to prescribing clopidogrel as a new genetic test offering. For each metric, a description of the metric, the calculated value that includes both quantitative data and qualitative descriptions, and their associated evidence, are provided. To provide realistic evidence, where possible, we drew from existing publications such as the Clinical Pharmacogenetics Implementation Consortium (CPIC) guidelines [24] (an international consortium facilitating the use of pharmacogenetic tests for patient care) and the PRECISE Value tool [25]. PRECISE Value is a prototype decision-analytic model (a decision tree) that estimates the value of developing and implementing clinical decision support (CDS) alerts for pharmacogenomics. [User Process #1, Tasks #4-6 – Rating pages]
- **Design consideration #3: Obtain and use GM service partner feedback.** For each metric included in the review form, potential GM service partners are

asked to rate the value and to provide their level of confidence in their assessment. They are also given an opportunity to add additional comments. Upon receiving multiple complete reviews, the system user can view an aggregate review and endorsement summary for a genetic test offering (Fig. 4). [User Process #2, Task #12 & 13 – Response summary and Reviewer summary]

D. Findings from End-user Studies

We conducted end-user sessions with seven GM service leaders from seven institutions. Six of them had experience serving on the Pharmacy and Therapeutics (P&T) committee (or committee with other names serving a similar institutional oversight function) and at least five of them have had a proposal for a new genetic test or practice. Overall, themes and points raised related to the two user processes and the three design considerations are summarized in Table I.

V. DISCUSSION

Our software design has the potential to improve the likelihood of implementing successful genetic test offerings, by facilitating key stakeholder review and endorsement of an offering. The level of endorsement for a genetic test offering may help GM service providers decide whether or not to move it through the institutional approval process. Our human-centered design strategy used a modified design sprint methodology. At the first step in our process, we found that taking time to reflect on findings from expert consultations helped us to revise our map to a solution. The revised map was then used to guide early mockups of the review and prioritization software and develop the low-fidelity prototype.

Across all tasks, user testing feedback indicated the following top three sub-themes: *add new information*, *general approval*, and *add/change functionality*. The second-rated sub-theme, *general approval* (48 comments, 4 synthesized comments) illustrates overall approval and satisfaction with the review and prioritization design and process. However, the other top-rated sub-themes *add new information* (92 comments, 14 synthesized comments) and *add/change functionality* (36 comments, 8 synthesized comments) identified areas of improvement to enhance the users' experience. All ten subthemes were relevant

Name	Specialty	Location	Email address	Select
Joe Doe	Oncology	Bayview Medical Center	joe1@jhmi.edu	☐
Stephan Doe	Neurosurgery	Johns Hopkins Hospital	stephan1@jhmi.edu	☐
Jenny Doe	Pediatrics	Sidney Kimmel Compre...	jenny1@jhmi.edu	☐
Joe Doery	Oncology	Johns Hopkins Hospital	joe2@jhmi.edu	☐
Stephan Doery	Neurosurgery	Bayview Medical Center	stephan2@jhmi.edu	☐

Fig. 2. Design consideration #1: Review & prioritization software prototype functionality to search for and select potential genomic medicine service partners.

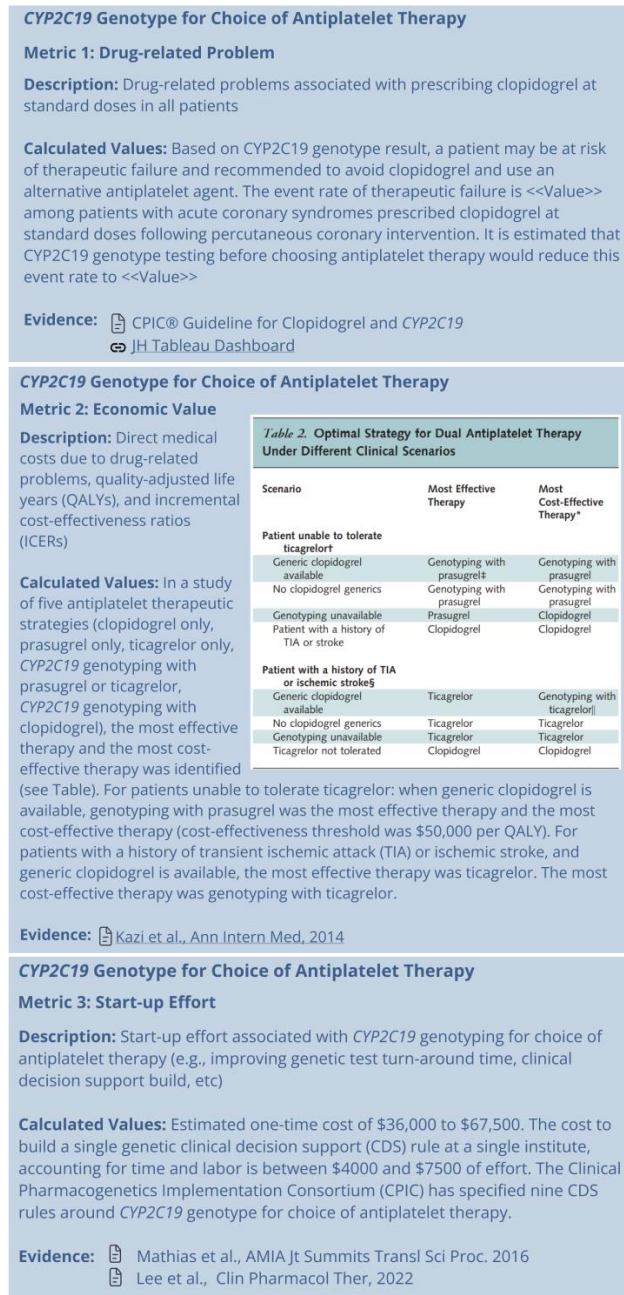


Fig. 3. Design consideration #2: Review & prioritization software prototype functionality to customize information about a genetic test offering being proposed. The information presented provides a summary of metrics that may be educational for potential genomic medicine service partners.

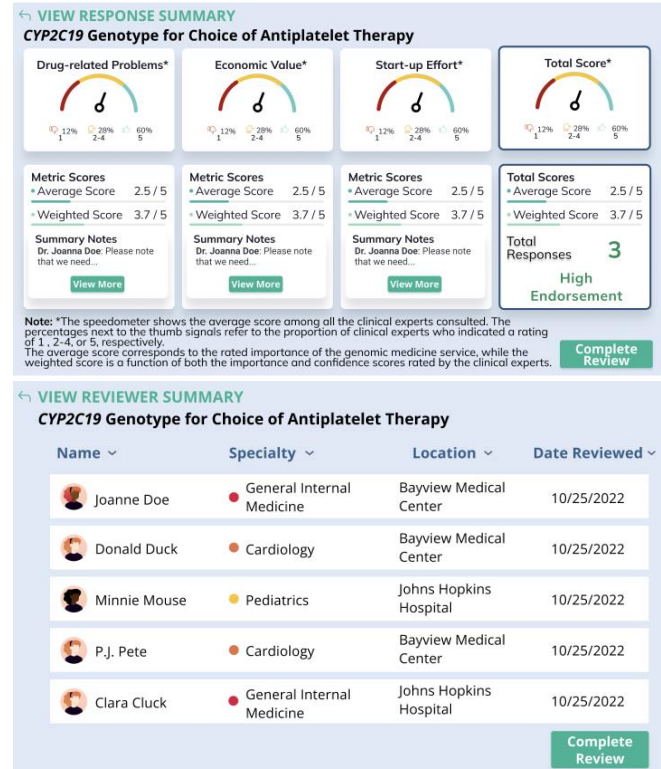


Fig. 4. Design consideration #3: Review & prioritization software prototype functionality to view an aggregate review and endorsement summary of a genetic test offering.

for user process #1 (create and distribute a review form to potential GM service partners), and eight were relevant for user process #2 (act upon review form responses). Given the focus of user process #2 on reviewing and acting on stakeholder feedback, it is not surprising that there were no synthesized comments for the *unclear of clinicians as targeted key stakeholders* and *add stakeholders* subthemes. Most of the synthesized comments were relevant to the three selected design considerations (61%, N=54). What follows describes insights gained from our review of those comments on prototype functionality supporting the three design considerations (Table I). We highlight areas that were viewed favorably by GM service providers, potential areas to improve, and the feasibility of making improvements.

A. Design Consideration #1: Identify Potential GM Service Partners to Champion a New Genetic Test Offering.

For this design consideration, we proposed functionality to search for and select potential clinical stakeholders from an institutional directory (Fig. 2). Findings from our sub-theme analyses, however, indicated that clinician stakeholders may not be sufficient to target as potential GM service partners. We found that the subthemes related to clinical stakeholders (*unclear of clinicians as targeted key stakeholders* and *add stakeholders*, 4 synthesized comments), subthemes providing positive feedback (*general approval* and *liked features*, 3

synthesized comments), and subthemes indicating a desire for more information (*add new information* and *more explanation needed*, 3 synthesized comments) were the most prevalent for this design consideration. We also had one synthesized comment under each of the *add/change functionality* and *software workflow issue* subthemes.

Findings from reviewing the *general approval* subtheme indicated that the design was visually appealing and that it had a clear and intuitive process. The subthemes related to clinical stakeholders indicated a desire for individuals beyond clinical providers to be included (e.g., pharmacists, lab professionals, and health IT professionals). From this feedback, we should expand our search capabilities beyond individuals with appointments in medical departments to also include other departments such as Pharmacy. The software provides flexibility for GM service leaders to select the individuals they believe are appropriate to review the evidence and to provide endorsement for a proposed genetic test offering.

The *add new information* subtheme highlighted a desire to include the credentials of individuals and whether they have genomic medicine or pharmacogenomics expertise. Other desired information was around the potential to engage or champion a new genetic test offering (e.g., an indicator of individuals who had been contacted for feedback previously and had completed a review). Such data could be tracked by the software and pulled into the search functionality. We are able to include more details about individuals, though there may be some details that would be missing by default. If implemented, these enhancements would help system users to make more informed stakeholder selections based on level of experience and areas of expertise. Feasibility to provide the most accurate and up-to-date information in requested areas, however, requires further investigation. Individuals, for example, may change career roles, employment, expertise, and location which can create challenges to maintaining such up-to-date information.

Finally, comments for *add/change functionality* and *software workflow issue* subthemes highlighted requirements for our move to a functional software. These included automating processes where possible (e.g., implementing typeahead, or auto-complete for searches), and making it easy for the user to save throughout the process (e.g., implement “save current progress” option that doesn’t require clicking a save button). Overall, we believe that these features and data access preferences (i.e., including individuals from more stakeholder groups and adding more details about those individuals) are feasible to implement. Doing so would allow potential users of the software to search and find potential GM service partners whose endorsement they believe will be critical for the success of a new genetic test offering.

B. Design Consideration #2: Train and Educate Stakeholders.

The review form that a GM service leader creates and distributes to selected stakeholders was used to implement this design consideration. Content included in the review form would both introduce stakeholders to a proposed genetic test offering and provide educational information about the service. For our study, we showed sample content in the prototype for one PGx genetic test offering: *CYP2C19* Genotype for Choice of

Antiplatelet Therapy. The content was presented under three metrics: drug-related problems, economic value, and start-up effort (Fig. 3). Similar to design consideration #1, we found several comments related to content-related subthemes (*add new information* and *confusing terminology*, 8 synthesized comments), subthemes that provide positive feedback (*general approval* and *liked features*, 6 synthesized comments), and subthemes related to clinical stakeholders (*unclear of clinicians as targeted key stakeholders* and *add stakeholders*, 3 synthesized comments). There were also comments under the *add/change functionality* subtheme (3 synthesized comments).

GM service leader interviewees provided positive feedback on the process to prepare content in the review form, on the information that was provided, the simplicity and conciseness of that information, and the visual characteristics of the form. There was also general approval of the resources we cited, including published articles and CPIC guidelines. The *add/change functionality* and content-related themes indicated a preference to add new information and some areas where the terminology used was confusing. Most areas for new information can be addressed with our current design since it is under user control (e.g., adding links to other evidence resources, more information on cost considerations and clinical impact, including more quantitative data about risk and cost-effectiveness, etc.). Other areas for new information and functionality could be supported by providing access to local data (e.g., interviewees indicated a desire for summaries on drug-related problems derived from electronic health record data). Feedback also indicated a need to improve how information is structured to improve readability and to enable quick assessment of a metric. Flexible format options should be supported to address this need. Furthermore, there was a need to support more customized content with one comment indicating a desire to edit and change what metrics are presented. The one *confusing terminology* comment raised could also be addressed by enabling more customization in the review form. The “calculated value,” for example, consistently added confusion, but there was no clear alternative label recommended. Last, the comments about clinicians as stakeholders relevant to this design consideration were about the appropriateness of the information for the audience and may influence what resources a GM service leader would reference.

Overall, there is a trade-off between providing comprehensive information and keeping that information brief and concise that should be studied further. With the added flexibility to send the review to a diversity of stakeholders (see Design Consideration #1), what and how information is presented may depend on the audience. The current design would give GM service providers control to trial-and-error what amounts and types of information are appropriate to include. An improved design would enable the user to make changes to field labels to better reflect the information provided and would provide functionality for flexible formatting options.

C. Design Consideration #3: Obtain and Use Genomic Medicine Service Partner Feedback

The third design consideration was addressed by the response summary page of the prototype (Fig. 4). That page provided a summary of genetic test review form submissions for intended

TABLE I. DESIGN CONSIDERATIONS AND SUB-THEMES

Process Step(s)	Sub-theme	Comment Count	Synthesized Comments & Design Considerations Design considerations #1, #2, and/or #3 related to the supporting note are indicated in parentheses ^a
(1), (2)	Add new information	92	- Frequency of the drug or phenotype (#3) - Dates and deadlines (#3) - Clinical stakeholder's credentials, PGx expertise, and feedback from their PGx review form (#1, #2, #3) - Visuals & Graphics (#3) - Inclusion of FDA, CPIC, and other PGx-related guidelines (#2) - Additional cost considerations (#2) - Clinical Impact of PGx such as adverse events (#2) - Numerical extraction of data associated with risk and cost-effectiveness data (#2) - Information about the patients and population (#2, #3) - Evidence and resources (#1, #2, #3) - More statistics and reporting (#3) - Demographics such as race/ancestry - Information about a genetic test offering - Time needed for CDS, and missing steps for the post-review clinical stakeholder process. (#3)
(1), (2)	General approval	48	- Clear page with relevant information and accessible resources (#1, #2, #3) - Good rating system and quantitative summary of responses (#3) - Clear and straightforward submission process (#2) - Organized method similar to but better than the institution's current process (#2, #3)
(1), (2)	Add/change functionality	36	- Accessibility to reviewer responses, and lead clinical champion (#3) - Automation of processes (#1) - Add free text and free space for adding rationale and notes/comments - Formatting and structuring data (#2,#3) - Ability to edit and change questions and the type of metrics (#2) - Functionality for additional resources (#2) - Design Aesthetics (#3) - Notifications and Alerts
(1), (2)	Liked feature	26	- Readability, simplicity, and brevity (#1, #2, #3) - Graphics and visuals (#1, #2, #3) - Links to resources (#2) - Ability to view specific dates and access entries based on the selected timeframe - Health system-specific PRECISE-Value modeling provides value for oversight committee
(1), (2)	Confusing terminology	26	- Main dashboard headers and scorecard terms - Unclear of the meaning of "calculated value", "genomic medicine service", the metric terms (#2) - PGx descriptions were too technical - Interchanging "view" and "create" for reviewing the form made the process unclear. - "Preview review form" and "review form" were too similar - The term "weighting" and "clinical experts' assignment was unclear" (#3)
(1), (2)	Software workflow issue	13	- Combine response summary and reviewer summary pages - Portal saving ability (#1) - Levels of endorsement not understood by user (#3) - The process was not easy for submission and overall review form creation; unclear dashboard info (#2) - Needed to reorder the review and prioritization process - Preview initial email sent to clinicians in the preview of the review form.
(1), (2)	Feature functionality is not intuitive	9	- Unclear functionality of certain features - Change design and indicate what the functionalities of certain features are
(1)	Unclear of clinicians as targeted key stakeholders	8	- Clinicians' opinions are not needed for review (#1,#2) - Unclear about reviewer or clinical experts' confidence and expertise, especially for the economic value metric (#1,#2)
(1), (2)	More explanation needed	8	- More informed guidance and a preview of the review form creation process - Explain score/weighted score information (#3) - Describe clinical champions background information (#1)
(1)	Add stakeholders	8	- Other medical professionals whose workflow will be impacted should also be consulted (including nurses, pharmacists, clinical laboratory, health IT) (#1, #2) - Emphasize the importance of identifying and selecting the key clinical champions who support the particular genetic test offering (#1)

^aNote: PGx = Pharmacogenomics, FDA = Food and Drug Administration, CPIC = Clinical Pharmacogenetics Implementation Consortium, GM = Genomic Medicine, CDS = Clinical Decision Support

use by GM service providers. Like the other design content-related subthemes (*add new information*, *confusing terminology*, and *more explanation needed*, 7 synthesized

comments) and positive feedback subthemes (*general approval* and *liked features*, 6 synthesized comments). There were also comments identified under *add/change functionality* (3 synthesized comments) and *software workflow issue* (1 synthesized comment) subthemes.

GM service provider interviewees indicated general approval of the response summary page design and described potential to improve current processes to summarize feedback from key stakeholders used at their institutions. Participants also liked the visuals and simplicity of the design. The *add/change functionality* and content-related subthemes, however, indicated room to improve the design. There was a particular desire for a way to view the evidence reviewed by stakeholders, to be able to see feedback from specific stakeholders, and for more quantitative data to be presented. In addition, there was specific feedback on design aesthetics including a desire for more visual cues like color schemes, charts, scales and legends in the response summary. The one *software workflow issue* subtheme comment was about a lack of clarity on how to interpret “endorsement” value. We intended for highly endorsed genetic test offerings to signal the user to move forward with its presentation to institutional governance groups such as the P&T committee. Such endorsement, however, may not be sufficient to prioritize a genetic test offering. The content-related comments shed some light on needs that GM service leaders may have beyond the stakeholder review summary. Comments indicated a desire for institution-specific details about the patient population, and frequency of drug or phenotype. Interviewees also wanted the ability to create a report that could be downloaded, printed, or shared electronically. These findings illustrated that GM service providers also need to compile local evidence for a new genetic test offering, and that evidence may be different from what is shared and reviewed by the selected stakeholders. Beyond the comments related to this design consideration, we found that the PRECISE-Value model was liked by most of our participants and may address this need for local evidence. To support decisions to move a genetic test offering forward, we should consider introducing such a model earlier in the user process. The current design introduced PRECISE-Value after the review summary. More work is needed to understand data access requirements to incorporate such a tool into the user process.

In sum, findings illustrated a need for designs such as what was shown in the response summary page is appropriate to support the work that GM service leaders do to synthesize feedback from stakeholders. We also found, however, that there is a need to improve our initial design so that the submissions made by stakeholders are more clearly summarized, and so that local evidence can be included in a report along with the stakeholder review summary. Doing this is a critical step toward our broader goal to enable prioritizing new genetic test offerings and decisions to move them through institutional approval processes.

D. Limitations and Future Work

First, we performed our analysis with a sample size of seven GM service leaders. While this is a small number, this is within a sample size that is common for qualitative, user experience, and participatory design research [26]. Second, many of the

synthesized comments were not relevant to the design considerations of focus in this work (39%, N=54). Other synthesized comments will be used to identify lower-priority design improvements. Third, we were unable to study the relationship between stakeholder engagement and organizational readiness for a GM service or genetic test offering. The relationship between stakeholder engagement and quality improvement projects is supported by empirical findings of others [27], but more work remains to empirically study this relationship with GM services specifically. Future work will study the relationship between stakeholder engagement and GM service and genetic test adoption. Furthermore, future work is required to establish the viability of our software design in the context of a range of GM services and genetic test offerings (including PGx and disease-focused GM services and genetic testing).

VI. CONCLUSION

We used a modified design sprint methodology to create prototype software for GM service leaders. The software would enable identifying key stakeholders, compiling evidence on a genetic test offering to distribute to those stakeholders, and would create a summary of feedback provided by stakeholders (a key stakeholder “response summary”). The response summary would enable GM service leaders to prioritize genetic test offerings to seek institutional approval for implementation. We also used an evaluation strategy that draws from the lessons of successful genomic medicine programs to identify and prioritize areas to improve the software in future design iterations. Our findings showed that the GM service leaders we interviewed generally approved of our software design and the proposed process to facilitate key stakeholders’ review and endorsement of a new genetic test offering. A few requirements and potential workflow issues were identified and will be prioritized to address in future iterations of the software design. High-priority areas include: expanding the types of stakeholders that can be reached beyond clinical providers, supporting features to customize the content being distributed to stakeholders, and exploring design alternatives for the response summary page to achieve optimal user understanding and clarity.

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